

Problem Discovery

In-Class Team Activity (and Presentation Guide)
DAEN 459 - Fall 2026

Goal for today: Clarify the problem before discussing development, architecture, or delivery.

Ground Rule: No Solutions Yet

For most of today's activity, do **not** discuss dashboards, databases, APIs, cloud platforms, machine learning models, visualizations, or implementation details.

If someone says, "We could build ...", ask:

What problem would that solve?

Outline

Time	Activity	Output
5 min	Initial problem statement	One sentence
5 min	Five Whys	Root problem
5 min	Stakeholder map	Stakeholder perspectives
5 min	Evidence vs. assumptions	Three critical assumptions
5 min	Failure even if the code works	Two discovery risks
5 min	Discovery questions	Key unknowns
5 min	Rewrite problem statement	Revised statement
5 min	Prepare report-out	90-second team presentation

1. Initial Problem Statement

Use this structure:

[Stakeholder] struggles with [problem] because [cause/context], resulting in [consequence].

Your current problem statement:

2. Five Whys

Start with your problem statement. Ask "Why does this matter?" several times until you reach an underlying need rather than a surface request.

Step	Why does this matter?
1	
2	
3	
4	
5	

What appears to be the deeper problem?

3. Stakeholder Map

Identify at least four stakeholder perspectives. Remember: the user, data owner, decision maker, and person affected by the output may be different people.

Stakeholder	What do they want?	What do they know?	What could go wrong for them?
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4. Evidence vs. Assumptions

Separate what you **know** from what you currently **assume**.

What we know	What we assume

Circle or list the three assumptions most likely to invalidate the project:

1. _____
2. _____
3. _____

5. How Could the Project Fail Even if the Code Works?

Generate at least five non-programming failure modes.

Examples:

- The data do not measure what we think they measure.
- The data are incomplete or unrepresentative.
- The wrong stakeholder defines success.
- Users do not trust or act on the output.
- The information arrives too late to support a decision.
- The output does not address the real problem.
- The problem is real but not important enough to justify the project.

1. _____
2. _____
3. _____
4. _____
5. _____

Our two most important discovery risks are:

1. _____
2. _____

6. Discovery Questions

Use the following prompts to identify what your team still needs to learn.

Problem

- What is happening today?
- Why is it a problem?
- How often does it occur?
- What is the consequence?
- What would happen if nobody solved it?

People

- Who experiences the problem?
- Who makes decisions related to it?
- Who owns or provides the relevant data?
- Who may be affected by the project's conclusions?

Data

- What evidence do we currently have?
- What data would we ideally want?
- What data actually exist?
- What important concepts might not be captured in the data?

Decisions

- What decision should become easier or better?
- Who makes that decision?
- What information do they currently use?
- How quickly must the information be available?

Unknowns

- What are we assuming?
- Which assumption is most dangerous?
- What do we need to learn before committing to this project?
- What evidence could convince us to change direction?

Our three most important unanswered discovery questions:

1. _____
2. _____
3. _____

7. Rewrite the Problem Statement

Use this structure:

For [stakeholder], [current situation/problem] creates [measurable consequence].
We need to understand [unknown/question] so that [decision or outcome] can be improved.

Revised problem statement:

8. Prepare the 90-Second Team Presentation — or at least, be prepared for it

Your report-out should focus on what you learned about the problem, not on what you plan to build.

Use this four-part structure:

1. **What we thought:** “We originally thought our project was about ...”
2. **What we learned:** “After discovery, we think the real problem is ...”
3. **What matters most:** “The most important stakeholder / assumption / risk is ...”
4. **What we still need to learn:** “Before building anything, we need to understand ...”

Presentation Constraint

Do not spend presentation time describing implementation, architecture, software tools, or proposed features.

The goal is to show that your team’s understanding of the **problem** became more precise.

Presentation Notes

1. **What we thought:**

2. What we learned:

3. Most important stakeholder, assumption, or risk:

4. What we still need to learn:

End-of-Class Deliverable: Discovery Canvas (fingers crossed!!)

Before leaving, your team should have concise answers for all six boxes below.

1. Problem	2. Stakeholders
3. Evidence	4. Assumptions
5. Unknowns	6. Revised Problem Statement

Follow-Up: Discovery Plan

For the next class or homework, identify:

- **3 questions** your team needs answered,
- **1 or 2 people** your team should talk to,
- **2 datasets or documents** your team should search, learn, or inspect,
- **1 assumption** your team should test,
- **1 reason** your team might substantially redefine or abandon the current project direction.